Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam F · Form 1: Advanced reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. From one level ball A is thrown upward at speed u; ball B is released from rest τ later. Both are airborne. After B is released, what are the acceleration and velocity of A relative to B (up positive)?

(A) 0; u-gτ

(B) -g; u

(C) -2g; u-gτ

(D) 0; u+gτ

2. A student walks 13 m right from a sensor, then 3 m left in a total of 6 s. Taking right as positive, which pair gives the additional displacement needed to return to the start and the total distance after returning?

(A) 10 m; 16 m

(B) Both zero

(C) +10 m; 16 m

(D) -10 m; 26 m

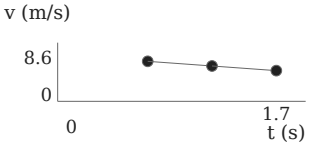
3. A bicycle moving right brakes uniformly. Use the model only until it stops. t (s) -> v (m/s): (0.7, 8.6); (1.2, 7.6); (1.7, 6.6). If sensor gain and physical slope are both unknown, what breaks their ambiguity?

(A) Calibrate the sensor at two independently known reference values.

(B) Repeat the same unknown setting.

(C) Assume any straight graph proves exact calibration.

(D) Reverse the ordering of the data.



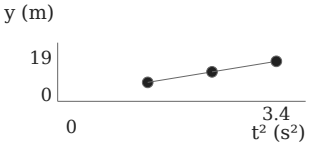
4. A sensor records downward position for a weight dropped from rest. Use g=10 m/s². Release from the same coordinate where gravity is halved. Plot the difference Ynew-Yold versus X. What slope should be observed?

(A) -1.45

(B) -4.6

(C) 0.65

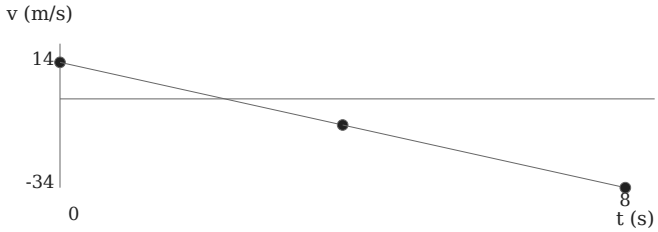
(D) -2.5



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A sensor records a cart with x0 = -7 m and v0 = +14 m/s. It has constant a = -6 m/s² for 8 s. Right is positive.

a. Describe the shapes and initial values of x–t, v–t, and a–t. **(1 pt)**



b. Find final position, displacement, and distance; explain any reversal. **(3 pt)**

6. A cart initially moving at -2 m/s receives a constant acceleration lasting 2 s each trial. a (m/s²) -> vfinal (m/s): (1.4, 0.8); (2.4, 2.8); (3.4, 4.8).

a. Build the model for velocity-area experiment from physics and predict both slope and intercept. **(2 pt)**

b. Double pulse duration without changing initial velocity. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A cart starts at x = 0 with initial velocity +u and constant acceleration -b, where u,b > 0. Derive the positive return time, round-trip distance, and average speed. **(3 pt)**